

1. A circle is inscribed in a unit square, and a diagonal of the square is drawn. Find the total length of the segments of the diagonal not contained within the circle.

Answer: $\sqrt{2} - 1$

Solution: The diagonal has length $\sqrt{2}$. It passes through the center of the circle, so it contains a diameter of the circle. The length of the diagonal within the circle is therefore 1. The rest of the diagonal outside of the circle has total length $\boxed{\sqrt{2} - 1}$.

2. Points D and E are chosen on the exterior of $\triangle ABC$ such that $\angle ADC = \angle BEC = 90^\circ$. If $\angle ACB = 40^\circ$, $AD = 7$, $CD = 24$, $CE = 15$, and $BE = 20$, what is the measure of $\angle ABC$ in degrees?

Answer: 70°

Solution: Since $\triangle ACD$ and $\triangle BCE$ are right triangles, we can use the Pythagorean Theorem to get

$$AC = \sqrt{7^2 + 24^2} = 25 \quad \text{and} \quad BC = \sqrt{15^2 + 20^2} = 25.$$

It follows that $\triangle ABC$ is isosceles, so

$$\angle ABC = \frac{180 - \angle ACB}{2} = \frac{180 - 40}{2} = \boxed{70^\circ}.$$

3. Let $ABCD$ be a convex quadrilateral with $BA = BC$ and $DA = DC$. Let E and F be the midpoints of BC and CD respectively, and let BF and DE intersect at G . If the area of $CEGF$ is 50, what is the area of $ABGD$?

Answer: 200

Solution: Let $[XYZ]$ denote the area of a polygon XYZ .

Observe that BF and DE are two medians of $\triangle BCD$, so G is the centroid of $\triangle BCD$. Since the medians of a triangle divide the triangle into six parts of equal area, we have

$$[CGF] = [CGE] = \frac{1}{2}[CEGF] = 25.$$

It follows that $[CBD] = 6 \cdot 25 = 150$ and $[BGD] = 2 \cdot 25 = 50$.

Furthermore, since $BA = BC$, $DA = DC$, and $BD = BD$, we have that $\triangle ABD \cong \triangle CBD$, so $[ABD] = [CBD] = 150$. Finally,

$$[ABGD] = [ABD] + [BGD] = 150 + 50 = \boxed{200}.$$

4. Let ABC be a right triangle with $\angle ABC$ right. Let D be a point on segment AC and let E be a point on ray AB past B such that $AC \perp DE$. Segment DE intersects segment BC at a point F such that $DF = FE$. Suppose that $\frac{CD}{BE} = 3$. Compute $\frac{AB}{AD}$.

Answer: $\frac{5}{3}$

Solution: Let $DF = FE = x$. Note that $\triangle CFD$ and $\triangle EBF$ are similar, with the ratio of the sides $\frac{CD}{BE} = 3$. Hence, $BF = \frac{1}{3}x$ and $CF = 3x$. Next, noting that $\triangle ABC$ is similar to $\triangle ADE$, we have $\frac{AB}{AD} = \frac{BC}{DE}$. Hence, the desired ratio is

$$\frac{AB}{AD} = \frac{BC}{DE} = \frac{BF + FC}{DF + FE} = \frac{\frac{1}{3}x + 3x}{x + x} = \boxed{\frac{5}{3}}$$

5. Plane A passes through the points $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$. Plane B is parallel to plane A , but passes through the point $(1, 0, 1)$. Find the distance between planes A and B .

Answer: $\frac{1}{\sqrt{3}}$ or $\frac{\sqrt{3}}{3}$

Solution: Plane A is described by the equation $x + y + z = 1$. The plane with this equation can easily be confirmed to pass through the three desired points, and planes are uniquely defined by three points. Since plane B is parallel to A , it has the same normal vector, but a different constant offset. Hence, B is described by $x + y + z = c$ for some c . Since it passes through $(1, 0, 1)$, we must have $c = 2$.

Next, we observe that the line $x = y = z$ is perpendicular to both planes. The distance between the planes can be found by the distance between the points of intersection of the line $x = y = z$ with both planes. Solving the system $x + y + z = 1$ and $x = y = z$ yields $x = y = z = \frac{1}{3}$, while solving the system $x + y + z = 2$ and $x = y = z$ yields $x = y = z = \frac{2}{3}$. Thus, the distance is

$$\sqrt{\left(\frac{2}{3} - \frac{1}{3}\right)^2 + \left(\frac{2}{3} - \frac{1}{3}\right)^2 + \left(\frac{2}{3} - \frac{1}{3}\right)^2} = \sqrt{\frac{1}{9} + \frac{1}{9} + \frac{1}{9}} = \boxed{\frac{1}{\sqrt{3}}} = \boxed{\frac{\sqrt{3}}{3}}.$$

6. Let ABC be a triangle with $AB = 5$ and $AC = 4$. Let D be the reflection of C across AB , and let E be the reflection of B across AC . D and E have the special property that D, A, E are collinear. Finally, suppose that lines DB and EC intersect at a point F . Compute the area of $\triangle BCF$.

Answer: $\frac{30}{7}\sqrt{3}$

Solution: Let $[XYZ]$ denote the area of a polygon XYZ .

Since $\angle DAB = \angle BAC = \angle CAE$, we must have that the three angles are 60° . We use this fact to compute the area of $\triangle ABC$, which is $\frac{1}{2} \cdot 5 \cdot 4 \sin 60^\circ = 5\sqrt{3}$. In addition, we compute using the Law of Cosines that BC^2 is

$$BC^2 = 5^2 + 4^2 - 2 \cdot 5 \cdot 4 \cos 60^\circ = 25 + 16 - 20 = 21.$$

Next, we compute the area of $\triangle BCF$ by finding the area of $\triangle DEF$ and subtracting away known areas; in this case, three times the area of $\triangle ABC$, since $\triangle ABC \cong \triangle ABD \cong \triangle AEC$. Note that since $\angle FDE = \angle ACB$ and $\angle FED = \angle ABC$, $\triangle ABC$ and $\triangle FED$ are similar. By using the fact that the ratio of the areas of a triangle is equivalent to the ratio of the side lengths, squared, the area of $\triangle DEF$ is

$$[DEF] = [ABC] \cdot \frac{(5+4)^2}{21} = 5\sqrt{3} \left(\frac{81}{21}\right) = \frac{135}{7}\sqrt{3}.$$

Finally, the area of $\triangle BCF$ is

$$[BCF] = [DEF] - 3[ABC] = \frac{135}{7}\sqrt{3} - 3 \cdot 5\sqrt{3} = \boxed{\frac{30}{7}\sqrt{3}}.$$

7. A circle intersects the y -axis at two points $(0, a)$ and $(0, b)$ and is tangent to the line $x+100y = 100$ at $(100, 0)$. Compute the sum of all possible values of $ab - a - b$.

Answer: 10000

Solution: Let $n = 100$ and define the following points: $A = (0, a)$, $B = (0, b)$, $C = (0, 1)$, and $D = (n, 0)$. By Power of a Point, $CA \cdot CB = CD^2$. Using the distance formula,

$$CD^2 = n^2 + 1^2 = n^2 + 1.$$

If the circle lies below the line, we have that $CA = 1 - a$ and $CB = 1 - b$. Then

$$(1 - a)(1 - b) = n^2 + 1 \implies ab - a - b = n^2 = 100^2 = 10000.$$

If the circle lies above the line, we have that $CA = a - 1$ and $CB = b - 1$, so we similarly get

$$(a - 1)(b - 1) = n^2 + 1 \implies ab - a - b = n^2 = 100^2 = 10000.$$

It follows that the only possible value of $ab - a - b$ is $\boxed{10000}$.

8. In rectangle $ABCD$, point E is chosen on AB and F is the foot of E onto side CD such that the circumcircle of $\triangle ABF$ intersects line segments AD and BC at points G and H respectively. Let S be the intersection of EF and GH , and T the intersection of lines EC and DS . If $\angle SFT = 15^\circ$, compute the measure of $\angle CSD$.

Answer: 105°

Solution: Since F is on the circumcircle of rectangle $ABHG$, we have that

$$\angle FDS = \angle FGH = \angle FBH = \angle FEC,$$

which implies that $FTED$ is cyclic. Since $\angle EFD = 90^\circ$, we have that $\angle ETD = 90^\circ$, so S is the orthocenter of $\triangle CDE$.

With this, we have that $CTSF$ is cyclic, so $\angle TFC = \angle TSC$. But

$$\angle TSC = \angle TFC = 90 - \angle SFT = 90 - 15 = 75,$$

so

$$\angle CSD = 180 - \angle TSC = 180 - 75 = \boxed{105^\circ}.$$

9. In quadrilateral $ABCD$, $AC = 5$, $CD = 7$, and $AD = 3$. The angle bisector of $\angle CAD$ intersects CD at E . If $\angle CBD = 60^\circ$ and $\angle AED = \angle BEC$, compute the value of $AE + BE$.

Answer: 8

Solution: First, we can use Law of Cosines to determine $\angle CAD$:

$$\cos \angle CAD = \frac{CD^2 - AC^2 - AD^2}{-2AC \cdot AD} = \frac{7^2 - 5^2 - 3^2}{-2(5)(3)} = \frac{15}{-30} = -\frac{1}{2}.$$

Hence, $\angle CAD = 120^\circ$.

Let B' be the image of B reflected over CD . Then $\angle CB'D = \angle CBD = 60^\circ$, so

$$\angle CAD + \angle CB'D = 120^\circ + 60^\circ = 180^\circ,$$

implying that $ACB'D$ is cyclic. But $\angle B'EC = \angle BEC = \angle AED$, so A , E , and B' are collinear. It follows that

$$\angle B'CD = \angle B'AD = 60^\circ \quad \text{and} \quad \angle B'DC = \angle B'AC = 60^\circ,$$

so $\triangle B'CD$ is equilateral with $B'D = B'C = CD = 7$.

Applying Ptolemy's Theorem on $ACB'D$, we have $AC \cdot B'D + AD \cdot B'C = CD \cdot AB'$. Substituting our known values, we have $5 \cdot 7 + 3 \cdot 7 = 7AB'$, so $AB' = 8$. As a result,

$$AE + BE = AE + B'E = AB' = \boxed{8}.$$

10. Circle ω_1 has diameter AB , and circle ω_2 has center A and intersects ω_1 at points C and D . Let E be the intersection of AB and CD . Point P is chosen on ω_2 such that $PC = 8$, $PD = 14$, and $PE = 7$. Find the length of PB .

Answer: $49/2$ or 24.5

Solution: Let $[XYZ]$ denote the area of a polygon XYZ .

Note that E is the midpoint of CD , so let $CE = ED = x$. We can use Stewart's Theorem on triangle PCD and cevian PE to get

$$\begin{aligned} 14^2x + 8^2x &= 7^2 \cdot (2x) + x \cdot x \cdot (2x) \\ \implies 196 + 64 &= 98 + 2x^2 \\ \implies x^2 &= 91, \end{aligned}$$

so $CE = ED = 9$. Using Heron's Formula, we can find the area of PCD . The semiperimeter is $\frac{1}{2}(14 + 18 + 8) = 20$, so the area is

$$\sqrt{20(20-14)(20-18)(20-8)} = 24\sqrt{5}.$$

The radius of ω_2 is then

$$\frac{PC \cdot PD \cdot CD}{4 \cdot [PCD]} = \frac{8 \cdot 14 \cdot 18}{4 \cdot 24\sqrt{5}} = \frac{21}{\sqrt{5}}.$$

Moreover, we can compute AE using the Pythagorean Theorem, giving us

$$AE = \sqrt{AC^2 - CE^2} = \sqrt{\frac{441}{5} - 81} = \frac{6}{\sqrt{5}}.$$

Since $\triangle ACE \sim \triangle ABC$, we have that $AE \cdot AB = AC^2 = AP^2$, so $\triangle APE \sim \triangle ABP$ by SAS. From this, we obtain

$$\frac{PE}{PB} = \frac{AE}{AP} = \frac{\left(\frac{6}{\sqrt{5}}\right)}{\left(\frac{21}{\sqrt{5}}\right)} = \frac{2}{7}.$$

Thus, $PB = \boxed{\frac{49}{2}}.$